

MILANO LINATE, Italy

WMO: 160800

Lat: 45.445N

Lon: 9.277E

Elev: 353

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 1999-2023

Grade: A

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.8	(c) 27.3	(d) 11.6	(e) 10.1	(f) 41.1	(g) 16.3	(h) 12.7	(i) 39.2	(j) 18.0	(k) 51.3	(l) 13.9	(m) 48.5	(n) 1.5	(o) 210	(p) 0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	92.9	74.4	90.5	73.0	88.3	71.7	77.2	88.2	75.3	86.4	73.7	84.8	4.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	127.7	83.6	71.8	119.2	81.4	70.1	112.2	80.0	40.7	87.9	38.8	86.2	37.3	84.6	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.2	11.1	9.1	DB	20.6	96.3	3.6	2.2	18.0	97.9	15.9	99.1	13.8	100.3	11.2	101.9	
			WB	19.3	78.6	3.8	2.5	16.6	80.4	14.4	81.9	12.3	83.3	9.6	85.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures and Degree-Days	DBAvg	58.3	38.7	42.3	50.2	57.5	65.9	74.2	77.6	76.5	68.9	59.1	48.3	39.7
	DBStd	14.78	4.78	5.57	5.93	5.52	5.64	5.25	4.13	4.75	5.00	5.69	5.74	5.47
	HDD50	1069	352	219	69	7	0	0	0	0	0	4	97	321
	HDD65	3704	816	635	459	232	57	3	0	0	21	196	501	784
	CDD50	4107	1	4	76	230	494	725	855	821	567	286	46	2
	CDD65	1265	0	0	0	6	86	277	390	356	137	13	0	0
Wind	WSAvg	3.6	3.0	3.6	4.1	4.5	4.4	4.3	4.2	3.5	3.2	2.7	2.8	2.8
Precipitation	PrecAvg	38.2	3.0	2.6	3.2	3.1	3.7	2.8	2.4	3.3	2.8	4.1	3.9	2.3
	PrecMax	55.6	10.8	5.9	8.0	9.6	11.4	6.0	8.0	7.0	8.4	16.4	10.4	6.1
	PrecMin	27.1	0.0	0.0	0.2	0.2	0.2	0.9	0.6	0.0	0.1	0.1	0.0	0.0
	PrecStd	8.9	2.6	1.8	2.1	2.2	2.8	1.3	1.9	2.2	2.3	3.7	2.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	64.2	74.1	79.6	88.7	94.4	95.3	96.8	87.7	79.1	65.2	57.7
		MCWB	46.8	49.5	56.3	60.9	69.8	76.2	73.4	76.3	69.9	64.6	57.9	48.3
	2%	DB	52.6	58.8	68.8	75.5	84.3	91.4	92.4	92.9	84.8	74.7	61.3	52.9
		MCWB	44.7	48.1	54.2	59.2	67.0	73.6	73.9	74.6	68.9	64.0	56.0	48.4
	5%	DB	49.1	55.4	65.1	72.4	80.8	88.5	90.2	90.1	82.4	71.7	58.8	50.3
		MCWB	43.8	46.2	53.0	57.8	65.2	71.5	73.1	73.0	67.6	62.6	55.1	46.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	52.5	59.4	63.7	71.5	79.3	79.0	80.9	73.9	67.9	60.8	52.3
		MCDB	53.9	60.3	69.0	75.0	86.4	92.2	90.4	93.4	83.9	73.2	62.3	54.4
	2%	WB	47.3	49.8	56.4	61.2	68.8	75.9	76.5	77.4	71.3	66.2	57.9	49.6
		MCDB	49.6	55.9	64.6	72.1	80.8	87.1	88.1	87.9	80.4	72.0	59.7	51.4
	5%	WB	45.2	48.1	54.3	59.5	66.8	73.6	75.1	75.2	69.6	64.5	56.1	48.0
		MCDB	47.7	52.7	62.3	69.4	77.8	85.0	86.7	85.6	78.5	70.4	58.0	49.4
Mean Daily Temperature Range	5% DB	WB	43.3	46.4	52.6	57.8	65.0	71.6	73.6	73.5	68.1	62.7	54.2	46.2
		MCDB	45.8	50.5	60.0	66.6	75.2	82.7	84.9	84.0	76.2	67.4	56.2	47.7
		MCWB	12.9	15.7	18.1	17.7	18.0	18.6	18.9	18.6	17.7	14.3	11.6	11.3
	5% WB	MCDBR	18.5	22.6	25.0	23.6	22.9	21.6	20.7	21.6	21.0	19.0	15.0	14.8
		MCWBR	12.4	13.9	12.9	10.7	10.0	9.4	8.5	8.2	8.3	9.9	9.8	10.6
		MCDBR	14.3	19.4	20.7	20.2	20.6	20.4	19.1	20.1	18.6	16.3	10.9	10.2
Clear-Sky Solar Irradiance	taub	0.331	0.366	0.405	0.458	0.464	0.487	0.472	0.462	0.450	0.432	0.377	0.332	
	taud	2.474	2.345	2.206	2.072	2.113	2.091	2.159	2.181	2.184	2.191	2.349	2.487	
	Ebn at Noon	245	253	258	253	256	250	252	250	240	224	220	231	
	Edh at Noon	22	30	39	49	48	50	46	43	40	35	25	20	
All-Sky Solar Radiation	RadAvg	473	731	1164	1491	1785	2042	2112	1777	1329	801	469	377	
	RadStd	64	111	137	162	140	120	132	102	87	111	77	56	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(44) Station	Trend	+0.52	+1.73	N/S	N/S	-1.26	-1.94	N/S	N/S	N/S	N/S	N/S
(45) Station	Variability	0.9	2.3	3.4	2.0	2.2	2.5	212	281	221	175	
(46) Regional Trend (19 neighbors)		+1.01	+1.75	N/S	+1.57	N/S	N/S	-148	-236	+185	+130	

Nomenclature: See separate page